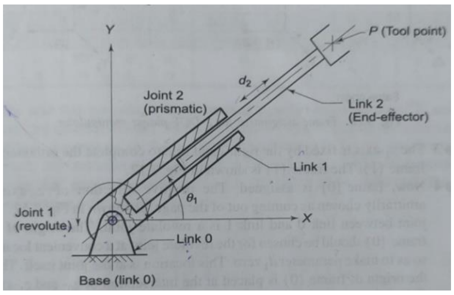
SASI INSTITUTE OF TECHNOLOGY & ENGINEERING
AUTONOMOUS
TADEPALLIGUDEM
WEST GODAVARI DISTRICT

Department of Mechanical Engineering

Academic Year: 2025-2026
MID EXAM-2 QUESTIONS

Course Name	Introduction Industrial Robotics	**Course Code**	23CAMEO606A
Branch & Semester	CSE & VI	**Date**	
Time	3:00PM TO 4:30PM	**Max. Marks**	30

Set-1

S. No	Questions	Marks	CO	BTL	PO & PSO
1	a) A 2DOF RP manipulator as shown in fig. below. 	5	3	3	PO1, PO2
	b) State the algorithm in Denavit-Hartenberg (D-H) notation	5	3	3	PO1, PO2
OR					
2	a) Consider two frames A and B. Frame B is obtained from frame A by a rotation of 30 degrees about the X-axis followed by a translation of 2 units along the Y-axis. Write the homogeneous transformation matrix that relates frame A to frame B.	5	3	3	PO1, PO2
	b) Find vector 3i+7j+5k for the translation matrix of, 3 units x, 5 units y and 6 units z directions.	5	3	3	PO1, PO2
3	a) It is desired to have the first joint of a 6-axis robot go from initial angle of 30 to a final angle of 75 _ in 5 seconds. Using a third-order polynomial, calculate the joint angle at 1, 2, seconds.	5	4	3	PO2, PO3
	b) A single cubic trajectory is given by $\theta(t) = 8+10t+45t^2+35t^3$ and is used over the time interval from t =1 to t = 2. What are the position, velocities and accelerations?	5	4	3	PO2, PO3
OR					
4	a) Differentiate between online and offline programming?	5	4	3	PO2, PO3
	b) Describe the relative merits and demerits of	5	4	2	PO2, PO3

	different textual robot languages.				
5	a) With a neat , explain the functions of machine vision?	5	5	2	PO1, PO4
	b) Explain the sensing function in machine vision	5	5	2	PO1, PO4
	OR				
6	a) Describe the digitization function in machine vision	5	5	2	PO1, PO4
	b) Explain the imaging devices with a neat sketches	5	5	2	PO1, PO4

Set-2

S. No	Questions	Marks	CO	BTL	PO & PSO
1	a) State the important steps in Denavit-Hartenberg (D-H) notation	5	3	3	PO1, PO2
	b) Solve the forward kinematics of 2DOF Robot RR manipulator?	5	3	3	PO1, PO2
	OR				
2	a) For the point (3 7 5) perform the following operations: a) Rotate 60^0 about X-axis b) Translate 9 units along y-axis c) Rotate 30^0 about x then translate 8 units along Y- axis.	5	3	3	PO1, PO2
	b) Solve the forward kinematics of 3DOF Robot RRR manipulator?	5	3	3	PO1, PO2
3	a) A single cubic trajectory given by $q(t)= 30+t^2-6t^3$ is used for a period of 3 seconds. Determine starting and final position, velocity and acceleration of end effector.	5	4	3	PO2, PO3
	b) Explain general terminology in trajectory planning?	5	4	2	PO2, PO3
	OR				
4	a) Describe the various methods used for path generation in robotic systems. What are the advantages and limitations of each method?	5	4	2	PO2, PO3
	b) A robotic manipulator is programmed to perform a pick-and-place operation involving skew motion.	5	4	2	PO2, PO3
5	a) Describe the lighting techniques in machine vision	5	5	2	PO1, PO4
	b) Explain the analog-to-digital conversion?	5	5	2	PO1, PO4
	OR				
6	a) Explain briefly image storage?	5	5	2	PO1, PO4
	b) Explain briefly image processing and analysis.	5	5	2	PO1, PO4

Set-3

S. No	Questions	Marks	CO	BTL	PO & PSO
1	a) State the kinematics relationship in Denavit-Hartenberg (D-H) notation	5	3	3	PO1, PO2
	b) Given the point $A_{uvw} = (4,2,7)T$ with respect to the rotated OUVW coordinate system, determine the coordinate system if it has been rotated 45^0 about the OX axis?	5	3	3	PO1, PO2
	OR				
2	a) Derive the homogeneous transformation matrix for a 2D rotation about (x, y).	5	3	3	PO1, PO2
	b) Find vector 25i+10j+20k for the translation matrix of, 8 units x, 5 units y and 0 units z directions.	5	3	3	PO1, PO2
3	a) Explain different methods of robot programming	5	4	2	PO2, PO3
	b) What are the steps in trajectory planning?	5	4	2	PO2, PO3
	OR				
4	a) Write down the capabilities and limitations of Lead through methods.	5	4	2	PO2, PO3
	b) Discuss the textural robot language structure with the help of block diagram	5	4	2	PO2, PO3
5	a) Describe the image data reduction system.	5	5	2	PO1, PO4
	b) Explain segmentation in machine vision	5	5	2	PO1, PO4
	OR				
6	a) Describe briefly object recognition system.	5	5	2	PO1, PO4
	b) Explain training the vision system	5	5	2	PO1, PO4

Set-4

S. No	Questions	Marks	CO	BTL	PO & PSO
1	a) Explain Joint space and World space coordinate systems	5	3	2	PO1, PO2
	b) Explain Forward and Inverse Kinematics	5	3	2	PO1, PO2
	OR				
2	a) Consider a 3-joint robot arm where the joint angles are θ1 = 45°, θ2 = 30°, and θ3 = 60°. The link lengths are L1 = 4, L2 = 3, and L3 = 2. Use forward kinematics to find the position of the end effector.	5	3	3	PO1, PO2
	b) Explain about homogeneous transformations in Robotics kinematics.	5	3	2	PO1, PO2
3	a) Explain briefly types of motion?	5	4	2	PO2, PO3

	b) Explain different Robot textual languages?	5	4	2	PO2, PO3
OR					
4	a) Differentiate between path planning and trajectory planning	5	4	3	PO2, PO3
	b) Explain the following: (i) Skew motion (ii) Straight line motion	5	4	2	PO2, PO3
5	a) Explain briefly inspection in robotic application.	5	5	2	PO1, PO4
	b) Explain briefly identification in robotic application	5	5	2	PO1, PO4
OR					
6	a) Explain briefly visual servoing and navigation in robotic application	5	5	2	PO1, PO4
	b) List advantages, disadvantages and applications of machine vision	5	5	1	PO1, PO4

Course Outcomes (COs)

CO1	**Identify** specific requirements of robots used for various Manufacturing applications in automated industries. **BTL-3 (Apply)**
CO2	**Characterize** various actuators and feedback sensors used for industrial robot applications. **BTL-3 (Apply)**
CO3	**Solve** forward and inverse kinematics problems using homogeneous transformations and D-H conventions. **BTL-3 (Apply)**
CO4	**Generate** safe and collision-free robot path using motion planning and control techniques. **BTL-3 (Apply)**
CO5	**Determine** appropriate machine vision techniques for robot inspection and automation applications. **BTL-3 (Apply)**

Signature of the Questions Paper Scrutiny

Course Coordinator	**Program Coordinator**	**Head of the Department**